

CHESSMASTER

"Challenge the Future, One Move at a Time"

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Goal

We created a chessboard with a robotic arm equipped with artificial intelligence and an online mode because we wanted to rediscover the authentic experience of playing chess. The idea comes from the desire to feel again that unique sensation of physically touching and moving the pieces on the chessboard, an aspect that traditional digital play often fails to convey.

The project allows you to play in two ways: alone, challenging the AI and going beyond the limits of traditional play, or with a friend at a distance thanks to the online mode.

In both cases, the pieces are still physically moved by the robotic arm, keeping the contact with reality alive and enriching the playing experience.

This combination of technology and physical interaction makes the game of chess more engaging and innovative, uniting tradition and modernity.

Structure

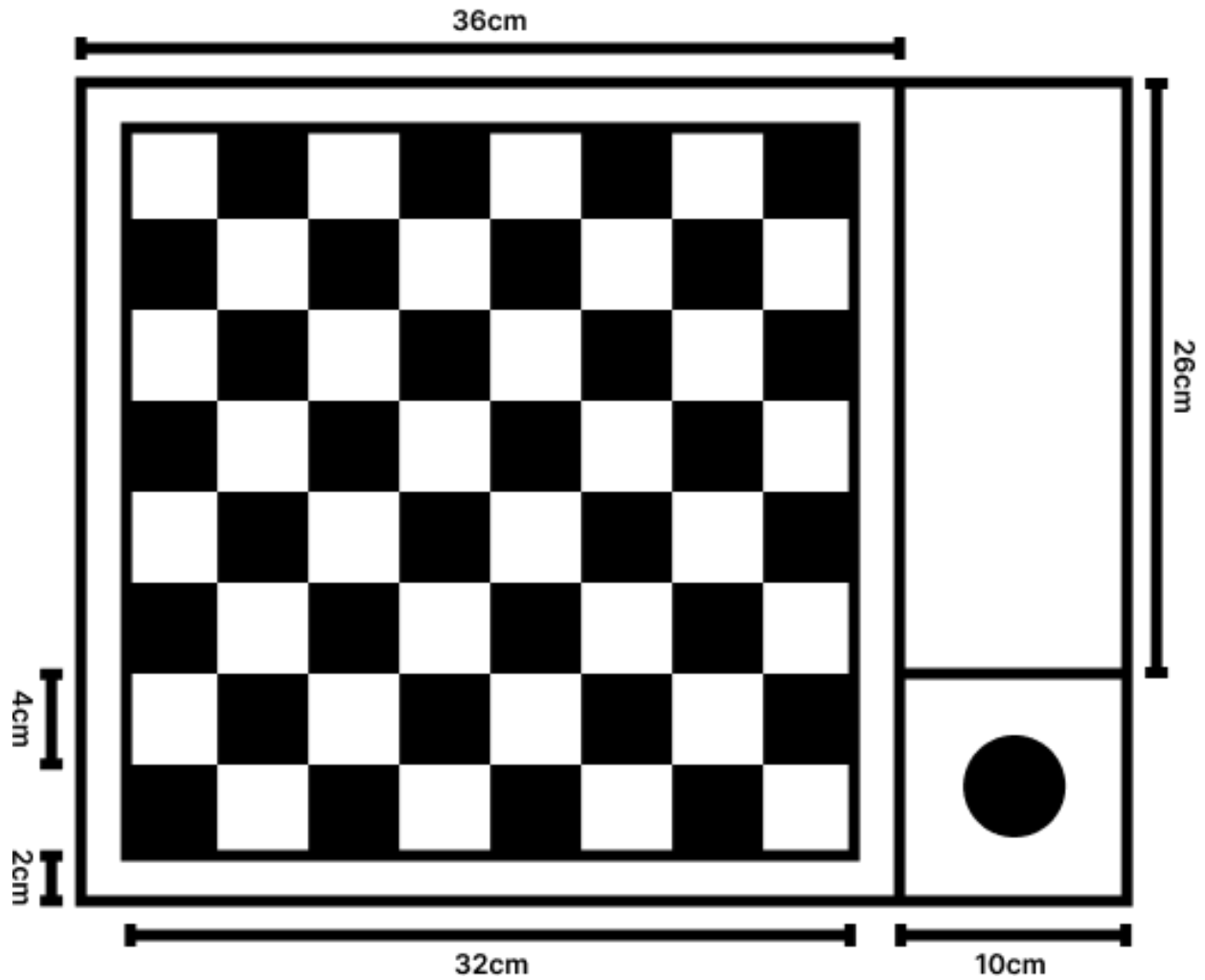
The Chessboard

The chessboard is built in plywood and is made up of two main parts: a front section and a rear section.

The front section, 7 cm high, includes the board, the joystick, and the piece pit. The board can be lifted, allowing quick access to the circuit in case problems occur with piece detection.

The outer surface is smooth, with the black squares colored with a Uniposca; inside, there are carved tracks that guide the positioning of the reeds. This section is connected to the rear part through folding hinges. The piece pit has a snap-on lid, and the wall separating the board from the joystick features a hole for cable passthrough.

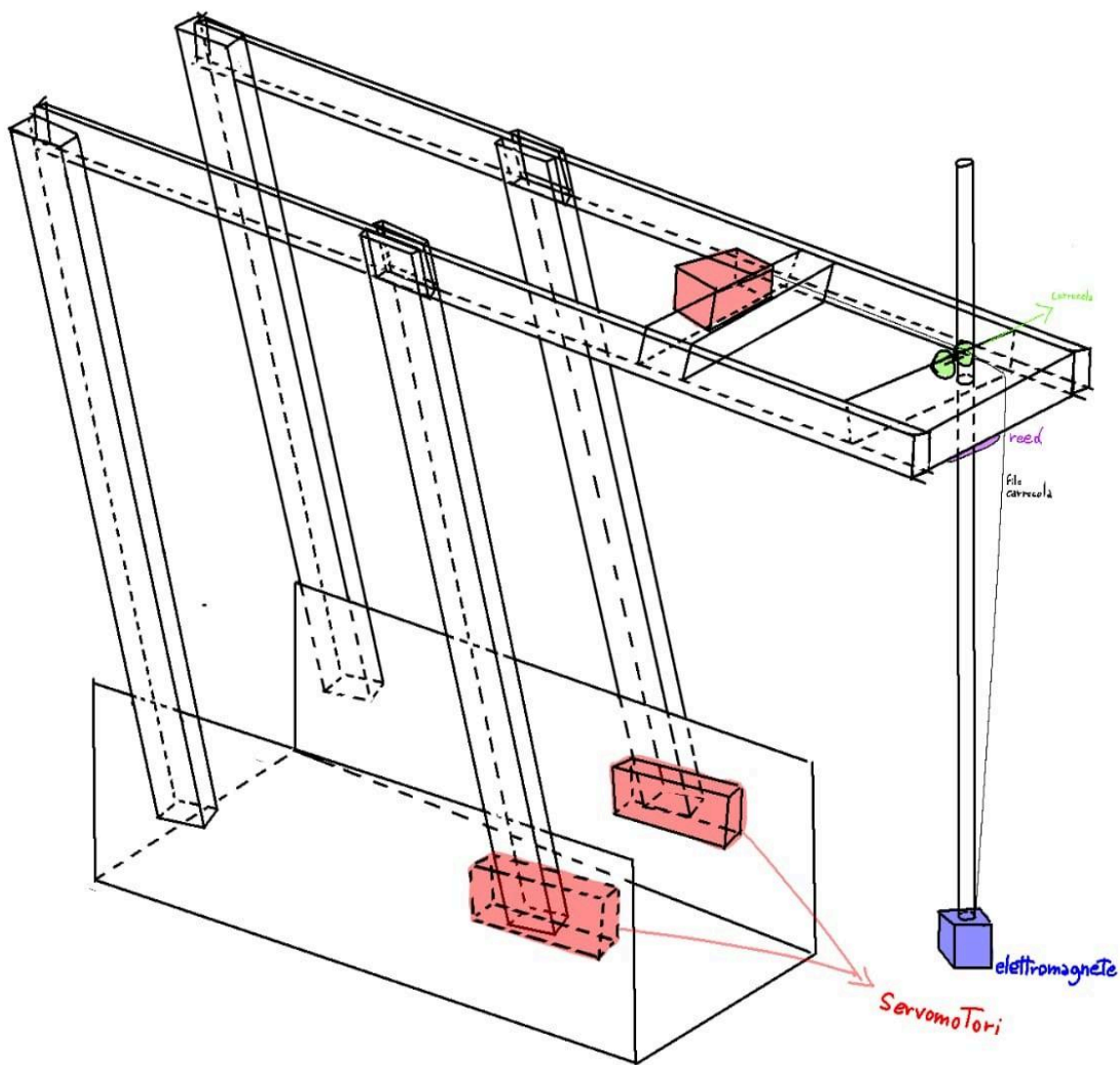
The rear part is a hollow parallelepiped measuring 15×46 cm and 20 cm high, directly connected to the front section. The wall joining the two parts contains a hole for the cables coming from the joystick and the board, as well as a hole that houses the display. The opposite wall can be opened and closed thanks to folding hinges and stays closed through magnets placed in the upper corners. On the left side are the hole for the power button and the one for connecting the power cable. The top panel contains a compartment for the servomotor, which allows the arm to rotate, and a dedicated hole for its cables. Inside the rear section are housed the Arduinos, the relay, the breadboard, and the entire circuit.



The Arm

The arm is fixed with screws to the servomotor mounted on the top surface of the rear part of the chessboard. It is made of a wooden base on which two servomotors are installed; these activate a four-bar linkage mechanism that allows the arm to move forward and backward. The two front rods are connected to the servomotors, while the rear ones are screwed to the base but can move freely. This system supports two additional rods joined together by two vertically mounted panels.

On the rear panel, an additional servomotor is fixed; it winds and unwinds the nylon thread, allowing the electromagnet to be raised and lowered. The front panel has two holes: through one passes a wooden cylinder with the electromagnet glued to its base, used to keep it always aligned with the chessboard; through the other passes the nylon thread, which is then tied to the electromagnet itself. Under the front panel, around the hole of the cylinder, there is a drilled wooden block that helps keep it perfectly vertical. A reed switch is also glued onto this block, and it is used to stop the upward movement of the mechanism when it approaches its limit.



Electronics

Piece Detection

The piece detection circuit works thanks to the magnets placed under each piece and the reed switches, small switches that react to magnetic fields. A reed switch is made of two metal blades enclosed in a small glass tube: when a magnet gets close, the blades attract each other and touch, closing the circuit; when the magnet moves away, the blades separate and the circuit opens.

The entire system is structured as an 8×8 matrix of reeds, for a total of 64 sensors, one for each square of the chessboard. Each reed closes when a magnetic piece is placed on the corresponding square. Rows and columns of the matrix are connected to the digital pins of an Arduino Mega 2560, following the indicated schematic. The columns include a $10\text{ k}\Omega$ pull-down resistor, so that the pins read a stable LOW signal when the column is not active.

Reading occurs through sequential scanning: Arduino activates one row at a time by setting it to HIGH, then checks which columns are also HIGH. If, for example, row 8 is active and columns 1, 2, and 3 read HIGH, it means that the corresponding reeds are closed and therefore squares A1, A2, and A3 are occupied. The microcontroller then moves to the next row until the entire chessboard is checked.

During the scan, Arduino records the values in an 8×8 matrix of 0s and 1s, where 0 indicates an empty square and 1 an occupied one. This method allows detecting all occupied positions using only 16 pins, instead of the 64 that would be required with one sensor per square. However, it only indicates whether a square is occupied, not which type of piece is occupying it.

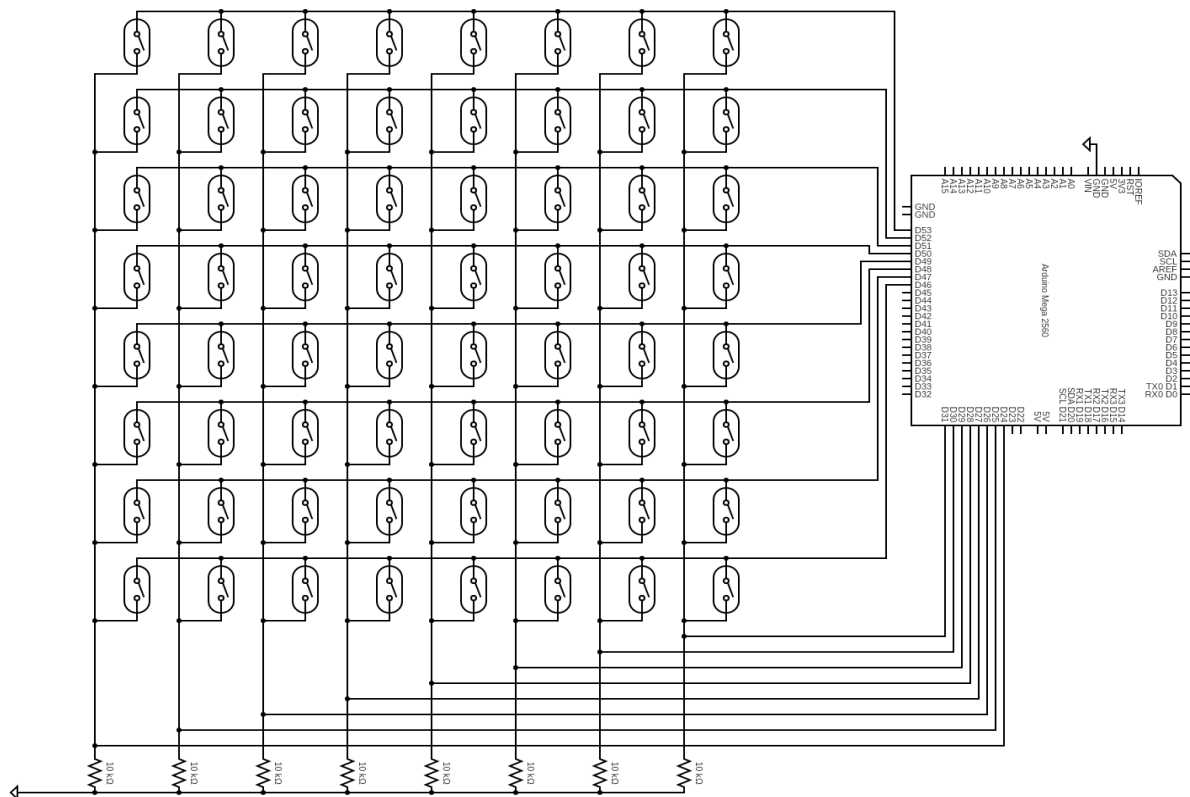
```

const int cell = 8;
int M[cell][cell];
for (int i = 0; i < cell; i++) {
    pinMode(cols[i], OUTPUT);
    pinMode(rows[i], INPUT_PULLUP);
}

for (int row = 0; row < cell; row++) {
    for (int col = 0; col < cell; col++) {
        for (int i = 0; i < cell; i++)
            digitalWrite(cols[i], LOW);

        digitalWrite(cols[col], HIGH);
        M[row][col] = digitalRead(rows[row]);
    }
}

```



Display and Joystick

The display and the joystick are used to select the game mode and modify some settings, allowing you to navigate inside the ChessOS in a simple and direct way.

Electromagnet

For piece collection, an electromagnet is used. Each piece has a small iron pin inserted on the top, which allows the magnet to lift it. The electromagnet has one terminal connected directly to the power supply, while the other is connected to the COM pin of a relay; the relay's NO pin is instead connected to the power supply's GND. Arduino controls the relay, thus turning the electromagnet on or off. A flyback diode is connected across the magnet's terminals (cathode on the positive, anode on the negative) to prevent interference on the rest of the circuit.

Go Up

Since the servomotor does not always perform the same stroke precisely, a reed switch was installed to signal Arduino when to stop the upward movement. On the rod that supports the electromagnet there is a small magnet: when the rod rises and reaches a certain height, the magnet activates the reed switch and Arduino stops the ascent.

Alimentation

The chessboard is powered by a desktop PC power supply. The +5V 18A line powers in parallel the Arduino Mega, the four servomotors, and the relay; the +12V 17A line is dedicated to the electromagnet, which requires more power. Through a breadboard, connected to the 5V and GND pins of the Arduino, the Arduino Wi-Fi, the display, and the joystick are also powered.

Software

The ChessMaster software is structured into three main components that work together to ensure an efficient and integrated operation of the chessboard.

This modular architecture allows for hardware resource optimization and ensures a scalable and robust management of the game, both locally and online.

Arduino Mega (Master)

This microcontroller acts as the main brain of the system. It manages the display's user interface through a dedicated operating system we call ChessOS. Here, all local operations are coordinated, such as displaying information and controlling the robotic arm and the pieces.

Arduino R4 WiFi (Slave)

This module is responsible for communicating with the external server. It acts as an intermediary, sending and receiving game data, moves, and status updates over the network, thus ensuring online connectivity and real-time synchronization.

API Server

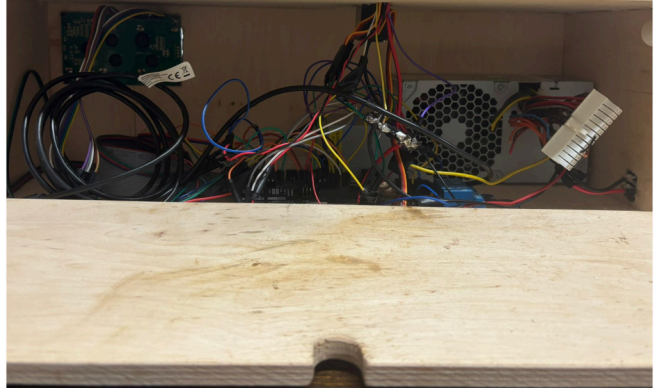
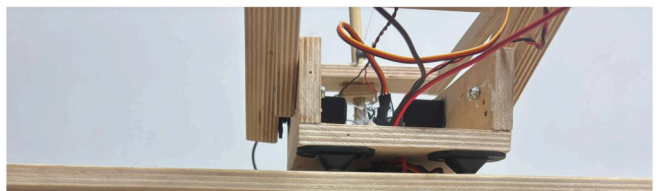
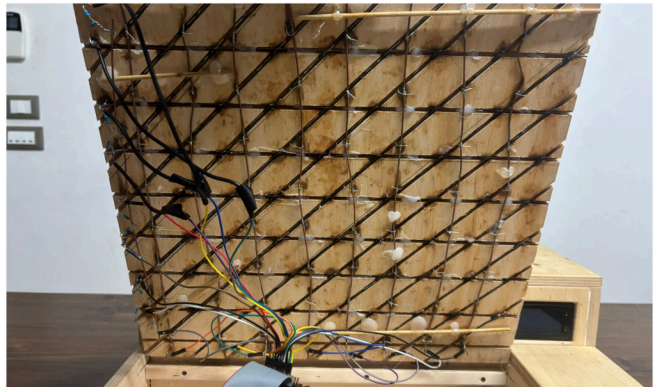
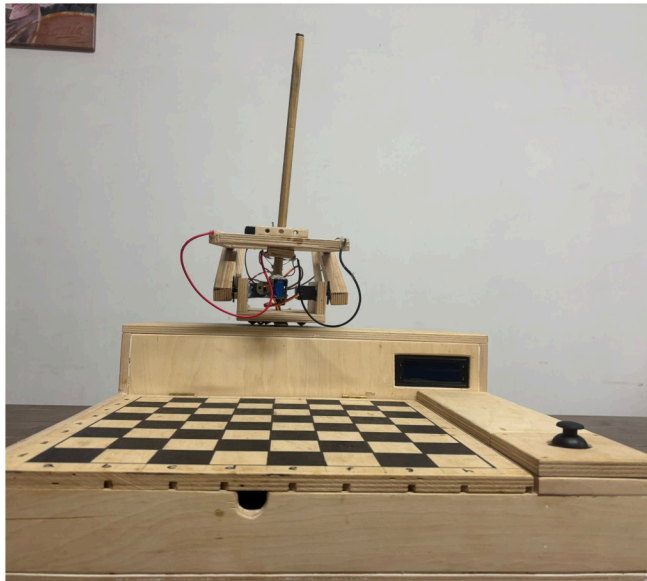
The remote server hosts the API with integrated WebSocket, which handles the game logic, uses the Stockfish chess engine for move processing, and coordinates connections between different players for online matches, ensuring a smooth and competitive gaming experience.

The source code is completely open source and can be viewed at the following link:

<https://github.com/raffreddat0/chessmaster>

Attachments

On the drive, it is possible to find more detailed photos, work-in-progress shots, and initial tests:
https://drive.google.com/drive/folders/1veMyuCROKsWigxhJ_fpz2VZzCziTvmm



Credits

We would like to express our sincere thanks to all the people who contributed with commitment and professionalism to the realization of this project.

Without their dedication and expertise, none of this would have been possible.

Structural Engineering

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